END-END COMMUNICATION AT TRANSPORT LAYER

AIM:

To implement echo client server using TCP/UDP sockets

TCP

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")

    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] Connection closed by client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Server received: {data}".encode())

    conn.close()
    server.close()

def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[client] Connected to server")

    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")

    client.close()

if __name__ == "__main__":
    server_thread = threading.Thread(target=start_server, daemon=True)
```

```
server_thread.start()
start_client()
```

OUTPUT:

```
[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to server
[SERVER] Connected by ('127.0.0.1', 53648) Enter message (or 'quit' to exit):
hello
[SERVER] Received: hello
[SERVER RESPONSE] Sever received: hello.
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.
```

UDP:

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the connection:")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped.")
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
```

```
            Print("[CLIENT] Closing connection ") break
        data, _ = c.recvfrom(1024)
        Print("Echo from server:", data.decode())
    c.close()
    Print("[Client] Stopped")
if __name__ == "__main__":
    threading.Thread(target=udp_server, daemon=True)
                .start() udp_client()
```

OUTPUT:

```
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000
                                                                    hi
[SERVER] Received from ('127.0.0.1', 49870): hi
Echo from Server: hi
Enter message (or 'quit' to stop): hello
[SERVER] Received from ('127.0.0.1', 49670): hello
Echo from Server: hello
Enter message (or 'quit' to stop): quit
[Client] Closing connection ... [SERVER] Client ('127.0.0.1', 4970
                                                    ended the connection.
[CLIENT] Stopped. [SERVER] Stopped.
```

RESULT:

Thus Implementing to echo client-server using TCP/UDP socket is done